

CLAYTON LAKE, ME, USA

WMO: 727119

Lat: 46.617N

Lon: 69.533W

Elev: 314

StdP: 97.61

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94605

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-29.1	-26.3	-35.8	0.1	-21.0	-32.2	0.2	-21.0	7.2	1.0	6.1	-2.6	0.5	280	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	28.5	19.3	27.0	18.4	25.5	17.7	21.2	25.6	20.3	24.4	19.4	23.6	3.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.9	15.1	22.6	18.9	14.2	22.4	17.8	13.2	21.9	62.8	25.9	59.7	24.1	56.7	23.5	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.4	4.7	-33.3	31.3	3.3	0.9	-35.7	31.9	-37.6	32.4	-39.5	32.9	-41.9	33.6		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-13.2	-10.8	-4.8	2.2	9.4	14.8	18.0	17.3	12.5	6.4	-0.7	-8.7
	DBStd	11.89	7.53	6.90	6.33	4.53	4.11	3.82	3.03	3.10	4.42	4.42	5.29	6.52
	HDD10.0	3115	718	583	460	236	61	4	0	0	22	131	322	579
	HDD18.3	5451	977	816	718	483	277	119	44	57	181	371	572	838
	CDD10.0	786	0	0	1	2	43	148	248	227	98	18	1	0
	CDD18.3	80	0	0	0	0	2	13	34	25	7	0	0	0
	CDH23.3	888	0	0	0	1	35	179	344	256	71	2	0	0
	CDH26.7	154	0	0	0	0	6	37	59	42	10	0	0	0
Wind	WSAvg	1.7	1.7	1.8	1.8	1.9	1.8	1.6	1.6	1.4	1.4	1.6	1.8	1.7
Precipitation	PrecAvg	1006	68	59	67	76	80	99	117	91	93	100	80	86
	PrecMax	1190	113	99	127	159	139	150	171	182	214	216	135	194
	PrecMin	793	13	28	29	30	28	33	60	4	52	25	44	40
	PrecStd	116	30	19	23	39	29	32	33	42	42	49	25	41
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	9.0	14.0	20.5	27.2	30.1	30.2	29.7	28.0	22.5	14.6	7.9
		MCWB	5.7	6.4	8.1	11.3	17.1	19.8	20.2	19.0	19.4	13.2	9.9	6.2
	2%	DB	3.8	4.9	8.8	16.1	23.6	27.5	28.4	27.8	25.4	19.0	11.6	4.4
		MCWB	2.6	2.5	3.4	9.3	14.6	18.3	19.4	18.8	18.1	11.7	7.8	3.2
	5%	DB	1.0	2.3	6.2	12.9	20.6	25.4	26.8	26.2	23.0	16.5	9.1	2.4
		MCWB	-0.3	0.5	1.8	7.1	13.0	17.0	18.6	17.9	16.9	10.8	6.5	1.1
	10%	DB	-1.5	0.5	4.1	10.0	18.1	23.1	25.3	24.6	20.9	14.1	7.0	0.9
		MCWB	-2.5	-0.8	1.0	5.2	11.6	15.7	17.8	17.7	15.2	9.3	5.1	-0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.3	6.8	10.1	14.1	19.3	21.4	22.8	21.8	21.3	17.5	11.9	6.3
		MCDB	6.2	8.7	13.0	19.1	23.8	27.5	27.0	26.8	25.4	20.1	13.7	7.5
	2%	WB	1.1	2.8	4.5	10.2	16.7	20.2	21.4	20.7	19.9	14.5	8.7	2.8
		MCDB	2.2	4.4	7.0	14.8	21.4	25.4	25.6	24.8	22.9	17.8	10.7	3.6
	5%	WB	-1.0	0.7	2.6	7.8	14.7	18.7	20.4	19.9	18.5	12.2	6.8	0.9
		MCDB	0.1	1.9	5.6	11.8	18.8	23.6	24.4	23.8	22.2	15.1	8.6	1.9
	10%	WB	-3.4	-0.9	1.0	5.8	12.8	17.3	19.4	18.9	16.9	10.2	5.0	-0.5
		MCDB	-2.5	0.6	3.7	9.8	17.2	21.9	23.7	23.3	20.6	13.2	6.9	0.5
Mean Daily Temperature Range	5% DB	MDBR	12.2	13.1	12.9	11.2	12.7	12.8	12.3	12.6	12.8	9.7	8.2	10.1
		MCDBR	11.5	13.0	15.5	16.9	17.3	16.2	14.3	14.8	14.8	14.3	11.2	9.0
		MCWBR	12.0	11.8	11.1	10.6	9.6	8.5	7.0	7.5	8.0	9.7	8.6	7.9
	5% WB	MCDBR	11.6	12.3	12.3	15.5	14.6	13.1	12.0	11.6	11.5	12.7	9.7	8.2
		MCWBR	11.2	11.2	9.2	10.7	8.9	7.5	6.3	6.0	6.7	9.4	8.4	7.8
Clear-Sky Solar Irradiance	taub	0.246	0.265	0.315	0.332	0.352	0.384	0.398	0.381	0.358	0.327	0.292	0.266	
	taud	2.386	2.312	2.254	2.411	2.426	2.328	2.306	2.363	2.418	2.505	2.527	2.408	
	Ebn at Noon	870	917	907	925	913	881	862	864	854	836	809	798	
	Edh at Noon	69	92	115	109	111	123	124	113	98	77	62	61	
All-Sky Solar Radiation	RadAvg	1.41	2.29	3.49	4.50	5.02	5.46	5.45	4.82	3.82	2.27	1.31	0.99	
	RadStd	0.18	0.27	0.23	0.39	0.54	0.35	0.48	0.29	0.28	0.20	0.14	0.09	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page